Geometric Signatures of Bayesian Inference in Discrete Dynamical Systems

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Abstract.

We introduce the I-predicate, a formal test for inference-like behavior in discrete dynamical systems on graphs, assembled from geometric criteria—progressive alignment to the final embedded state (measured via Kendall rank correlation), dimensional reduction, and data compression—measured under independent graph embeddings. We prove that any graph dynamical system whose embedded state trajectory converges to the posterior distribution of a well-defined latent variable model must satisfy the I-predicate (Theorem 3.1), with consecutive alignment rate bounded by the per-step information gain and participation ratio decreasing as $O(|\Theta|/t)$. The bound on progressive alignment follows from the Davis–Kahan/Wedin $\sin\theta$ theorem applied through the chain: posterior concentration (Doob) $\rightarrow$ graph convergence (smoothness of the encoding) $\rightarrow$ embedding convergence (regularity condition E1) $\rightarrow$ subspace alignment (spectral gap condition from E3). We also prove a partial converse: systems whose spectral gap evolution violates a quantitative monotonicity condition cannot satisfy the I-predicate (Theorem 5.1). As computational illustration, we construct four graph dynamical systems and verify that the I-predicate correctly classifies all four, including a betweenness-centrality contraction mapping that converges without Bayesian structure and is classified I-negative.

1 Introduction

1.1 Motivation

Recent work by Agarwal, Dalal, and Misra [2, 1] has revealed that transformer neural networks implement exact Bayesian posterior updating, and that this manifests as specific geometric signatures in the network's internal representations: progressive alignment of key-query subspaces, dimensional reduction of value manifolds, and directional flow on the Grassmannian. Zhang Chong [10] and the same authors have shown that these signatures appear identically in Mamba—a state-space model that uses no attention mechanism at all. This architecture-independence suggests the signatures may be properties of Bayesian inference itself, not of any particular computational substrate.

This raises a natural question: under what formal conditions does a discrete dynamical system on graphs provably exhibit these geometric signatures? And can the signatures be

assembled into a robust predicate—a formal test—that detects inference-like behavior in systems far removed from neural networks?

1.2 Contributions

1. **The I-predicate.** We define a formal predicate for inference-like behavior (Definitions 2.1–2.5, Section 2), assembled from geometric criteria measured under two independent embedding methods.

2. **Forward theorem.** We define precisely what it means for a graph dynamical system to "implement Bayesian posterior updating" (Definition 2.6) and prove that any such system satisfies the I-predicate (Section 4).

3. **Spectral gap obstruction (partial converse).** We prove that systems whose spectral gap sequence violates a monotonicity condition cannot be I^+ under the Laplacian embedding (Section 5).

4. **Discrimination examples.** We construct four graph dynamical systems and verify that the I-predicate correctly classifies all four (Section 6).

1.3 Scope and non-claims

We do not claim that graph dynamical systems "think," "learn," or "reason." The I-predicate detects specific measurable geometric properties; whether those properties constitute "inference" in any deeper sense is an interpretive question that this paper does not address.

2 Framework

2.1 Graph dynamical systems

Definition 2.1 (Graph dynamical system)**.** A graph dynamical system is a triple (G_0, R, T) where G_0 is an initial graph, R is a deterministic graph rewrite rule, and $T \in \mathbb{N}$ is a time horizon. The trajectory is the sequence $(G_0, G_1, \ldots, G_T)$ where $G_{t+1} = R(G_t)$.

Initial condition protocol. Each system is executed from a family of four canonical initial graphs $G_0 \in \{K_1, K_2, K_3, P_3\}$. A system is classified as I^+ only if it satisfies the I-predicate for at least three of the four initial graphs.

2.2 Embeddings and regularity conditions

Definition 2.2 (Embedding)**.** An embedding is a map $e : \{\text{graphs}\} \to \mathbb{R}^{n \times d}$ assigning to each graph G a matrix $X^e(G)$ whose rows correspond to nodes and columns to latent dimensions.

Standard embedding family. We fix two independent embedding methods $\mathcal{E} = \{e_L, e_R\}$:

- e_L **(Laplacian eigenvectors).** The columns of $X^{e_L}(G)$ are the d leading eigenvectors of the graph Laplacian $L(G) = D - A$.

- e_R **(Random projection).** $X^{e_R}(G) = A(G) \cdot M$ where $A(G)$ is the adjacency matrix and $M \in \mathbb{R}^{n \times d}$ is a fixed random Gaussian matrix.

Conditions E1–E3 (Embedding regularity).

- **E1 (Continuity).** If G and G' differ by a single edge, then $\|X^e(G) - X^e(G')\|_F \leq C_1$.
- **E2 (Non-degeneracy).** For any non-isomorphic graphs G, G', the column spaces $\mathrm{col}(X^e(G))$ and $\mathrm{col}(X^e(G'))$ are distinct points on the Grassmannian.
- **E3 (Spectral faithfulness).** The singular value distribution of $X^e(G)$ is Lipschitz in the spectral distribution of the graph Laplacian of G.

Verification: Laplacian eigenvectors satisfy E1 by Weyl's inequality [9], E2 by spectral characterization, and E3 by construction. Random projection satisfies E1 by the Johnson–Lindenstrauss lemma [5], E2 with probability 1, and E3 by concentration of projected singular values. See Appendix A.

2.3 The I-predicate

Definition 2.3 (Bayesian geometric signature)**.** Let X_T^e denote the embedding of the final graph. Define the cosine-to-final sequence as

$$c_t^e = \cos\theta\big(\mathrm{col}(X_t^e),\ \mathrm{col}(X_T^e)\big)$$

for $t = 0, \ldots, T-1$. A trajectory satisfies *progressive alignment* under embedding e if the Kendall rank correlation $\tau_{\text{to_final}}(e) = \tau(\{c_t^e\}, \{t\})$ exceeds $\tau^* = 0.5$. The trajectory satisfies *dimensional reduction* if $\mathrm{PR}(X_{t+1}^e - X_t^e)$ is monotonically non-increasing in t.

Definition 2.4 (Compression criterion)**.** The trajectory satisfies the compression criterion if its lossless compression ratio $\rho(\tau) < 0.85$.

Definition 2.5 (I^+)**.** A system is I^+ if: (i) it satisfies the compression criterion, AND (ii) it satisfies the Bayesian geometric signature under at least one embedding in $\mathcal{E}$, AND (iii) under no embedding does $\tau_{\text{to_final}}$ fall below $-\tau^*$ (anti-convergence guard). The system must satisfy the initial condition protocol.

2.4 Bayesian graph dynamical systems

Definition 2.6 (Bayesian graph dynamical system)**.** A graph dynamical system (G_0, R, T) implements Bayesian posterior updating with respect to a latent variable model $(\Theta, \{P_\theta\}, \pi_0)$ if there exists:

- A finite parameter space Θ with prior π_0,
- A likelihood model $P_\theta(G_t \mid G_{t-1})$ for each $\theta \in \Theta$,
- An encoding map $\phi : \Delta(\Theta) \to \{\text{graphs}\}$,

such that for all t: $G_t = \phi(\pi_t)$ where $\pi_t(\theta) \propto P_\theta(G_t \mid G_{t-1}) \cdot \pi_{t-1}(\theta)$, and the encoding ϕ satisfies:

- **Faithfulness:** ϕ is injective.
- **Smoothness:** If $\|\mu - \nu\|_{\mathrm{TV}} < \varepsilon$, then $\mathrm{d}_{\mathrm{edit}}(\phi(\mu), \phi(\nu)) \leq C_2 \cdot \varepsilon$.

3 Forward Theorem: Statement

3.1 Geometric quantities

- **Consecutive subspace cosine:** $\cos\theta(\mathrm{col}(X_t^e), \mathrm{col}(X_{t+1}^e))$.
- **Cosine-to-final:** $c_t^e = \cos\theta(\mathrm{col}(X_t^e), \mathrm{col}(X_T^e))$.
- **Kendall τ to final:** $\tau_{\text{to_final}}(e) = \tau(\{c_t^e\}, \{t\})$.
- **Participation ratio:** $\mathrm{PR}(M) = (\sum \sigma_i)^2/(n \cdot \sum \sigma_i^2)$.
- **Cumulative information gain:** $I_t = \sum_{s=1}^t D_{\mathrm{KL}}(\pi_s \| \pi_{s-1})$.

3.2 Statement

Theorem 3.1 (Forward theorem)**.** *Let (G_0, R, T) be a Bayesian graph dynamical system with respect to $(\Theta, \{P_\theta\}, \pi_0, \phi)$ in the sense of Definition 2.6, and let e be an embedding satisfying E1–E3. Let $K = |\Theta|$ and let $G_\infty = \phi(\delta_{\theta^*})$ be the limit graph. Suppose the singular value gap $\gamma_0 = \mathrm{gap}_k(X^e(G_\infty)) > 0$.*

Then there exists a warm-up time $t_0 = \min\{t : I_t > \log(4C_1C_2K/\gamma)\}$ with $\gamma = \gamma_0/2$, such that for all $t \geq t_0$:

(i) ***Progressive alignment.***

$$\cos\theta\big(\mathrm{col}(X_t^e),\ \mathrm{col}(X_{t+1}^e)\big) \geq 1 - C_3 \cdot e^{-2I_{t-1}}$$

where $C_3 = 2C_1^2C_2^2(K-1)^2/\gamma^2$.

(ii) ***Dimensional reduction.***

$$\mathrm{PR}(X_{t+1}^e - X_t^e) \leq C_4 \cdot K \cdot e^{-I_{t-1}}/n$$

where $C_4 = 4C_2$ and n is the number of nodes.

Corollary 3.2 ($\tau_{\text{to_final}}$ convergence)**.** *Under the hypotheses of Theorem 3.1, if $T > 4t_0 + 1$, then $\tau_{\text{to_final}}(e) \geq 1 - 2t_0/(T-1)$. In particular $\tau_{\text{to_final}} \to 1$ as $T \to \infty$.*

Corollary 3.3 (I-positivity)**.** *Any Bayesian graph dynamical system with sufficient cumulative information gain and sufficient time horizon satisfies the I-predicate under any embedding satisfying E1–E3, provided the trajectory compression ratio is below the threshold.*

Corollary 3.4 (Explicit I-positivity bound)**.** *Let (G_0, R, T) be a Bayesian graph dynamical system with parameters $(\Theta, \{P_\theta\}, \pi_0, \phi)$. Let $K = |\Theta|$, and define the minimum discrimination rate*

$$D_0 = \min_{\theta \neq \theta^*} D_{\mathrm{KL}}(P_{\theta^*} \| P_\theta) > 0.$$

Then the system is I^+ under any embedding satisfying E1–E3, provided:

$$T > 4\left(\frac{4C_1C_2(K-1)}{\gamma_0}\right)^{1/D_0} + 1.$$

If additionally the per-step KL divergence is bounded below by $\delta_0 > 0$, the stronger bound holds:

$$T > \frac{4}{\delta_0} \log\left(\frac{4C_1C_2(K-1)}{\gamma_0}\right) + 1.$$

Proof. By Lemma 4.1, $I_t \geq D_0 \log t$ for t sufficiently large. Therefore $t_0 \leq (4C_1C_2(K-1)/\gamma_0)^{1/D_0}$. By Corollary 3.3, I-positivity holds for $T > 4t_0 + 1$, giving the first inequality. For the strong-signal case: $I_t \geq \delta_0 t$ gives $t_0 \leq (1/\delta_0)\log(4C_1C_2(K-1)/\gamma_0)$, and $T > 4t_0 + 1$ gives the second inequality. □

Remark 3.5. The dominant cost is poor identifiability: the exponent $1/D_0$ shows that hard-to-distinguish hypotheses produce exponentially longer warm-up. In the strong-signal regime ($\delta_0 > 0$), the bound becomes logarithmic in the model parameters.

4 Forward Theorem: Proof

4.1 Notation

Throughout this proof, (G_0, R, T) is a Bayesian graph dynamical system with respect to $(\Theta, \{P_\theta\}, \pi_0, \phi)$. We write $K = |\Theta|$, $\pi_t \in \Delta(\Theta)$ for the posterior at time t, $G_t = \phi(\pi_t)$ for the encoded graph, and $X_t = X^e(G_t) \in \mathbb{R}^{n\times d}$ for the embedding under a fixed embedding e satisfying E1–E3. We write $\sigma_j(M)$ for the j-th singular value of M in decreasing order, $\mathrm{col}(M)$ for the column space, $d_{\mathrm{Gr}}(U, V)$ for the chordal distance on $\mathrm{Gr}(k, d)$, and $\theta_{\max}(U, V)$ for the largest principal angle between subspaces U, V.

4.2 Step 1: Posterior concentration (Doob)

Lemma 4.1 (Finite Doob consistency)**.** *Let Θ be finite with $|\Theta| = K$, let π_0 be a prior with $\pi_0(\theta) > 0$ for all θ, and let $\theta^* \in \Theta$ be the true parameter. Then for all $t \geq 1$:*

$$\|\pi_t - \delta_{\theta^*}\|_{\mathrm{TV}} \leq (K-1) \cdot e^{-I_t}$$

where $I_t = \sum_{s=1}^{t} D_{\mathrm{KL}}(\pi_s \| \pi_{s-1})$ is the cumulative information gain.

Proof. For each incorrect parameter $\theta \neq \theta^*$, the posterior satisfies $\pi_t(\theta) = \pi_0(\theta) \cdot \prod_{s=1}^{t}[P_\theta(G_s \mid G_{s-1})/P_{\theta^*}(G_s \mid G_{s-1})]$. By the standard martingale argument relating cumulative log-likelihood ratios to cumulative KL divergence [4], $\pi_t(\theta) \leq \pi_0(\theta) \cdot e^{-I_t}$ for each $\theta \neq \theta^*$. Summing over $\theta \neq \theta^*$ gives the bound. □

Corollary 4.2. *The consecutive TV distance satisfies $\|\pi_t - \pi_{t-1}\|_{\mathrm{TV}} \leq 2(K-1) \cdot e^{-I_{t-1}}$, by the triangle inequality through δ_{θ^*}.*

4.3 Step 2: Graph convergence

Lemma 4.3. *Under the smoothness of ϕ (Definition 2.6), for all $t \geq 1$:*

$$\mathrm{d}_{\mathrm{edit}}(G_t, G_{t+1}) \leq C_2 \cdot \|\pi_t - \pi_{t+1}\|_{\mathrm{TV}} \leq 2C_2(K-1) \cdot e^{-I_{t-1}}.$$

Proof. Direct from the smoothness condition and Corollary 4.2. □

4.4 Step 3: Embedding convergence

Lemma 4.4. *Under E1, for all $t \geq 1$:*

$$\|X_{t+1} - X_t\|_F \leq C_1 \cdot \mathrm{d}_{\mathrm{edit}}(G_t, G_{t+1}) \leq 2C_1C_2(K-1) \cdot e^{-I_{t-1}}.$$

Proof. Each single-edge edit changes the embedding Frobenius norm by at most C_1 (E1). The edit distance counts the minimum number of single-edge operations, so iterating E1 and applying the triangle inequality gives $\|X^e(G') - X^e(G)\|_F \leq C_1 \cdot \mathrm{d}_{\mathrm{edit}}(G, G')$. Combined with Lemma 4.3. □

Write $\varepsilon_t = \|X_{t+1} - X_t\|_F$ for the embedding perturbation at step t. By Lemma 4.4, $\varepsilon_t \leq 2C_1C_2(K-1) \cdot e^{-I_{t-1}}$.

4.5 Step 4: Subspace alignment (Davis–Kahan)

4.5.1 The singular value gap condition

Proposition 4.5 (E3 implies gap non-collapse)**.** *Let $G_\infty = \phi(\delta_{\theta^*})$ be the limit graph. If e satisfies E2 and E3, then $\mathrm{gap}_k(X^e(G_\infty)) = \gamma_0 > 0$. Define $t_0 = \min\{t : I_t > \log(4C_1C_2(K-1)/\gamma_0)\}$. Then for all $t \geq t_0$:*

$$\mathrm{gap}_k(X_t) \geq \gamma := \gamma_0/2.$$

Proof. By Weyl's inequality [9], $|\sigma_j(X_t) - \sigma_j(X^e(G_\infty))| \leq \|X_t - X^e(G_\infty)\|_F$ for each j. By Lemmas 4.1 and 4.4, $\|X_t - X^e(G_\infty)\|_F \leq C_1C_2(K-1)e^{-I_t}$. The definition of t_0 ensures this is less than $\gamma_0/4$ for $t \geq t_0$. Since the gap can decrease by at most $2\|X_t - X^e(G_\infty)\|_F$, we get $\mathrm{gap}_k(X_t) \geq \gamma_0 - 2 \cdot \gamma_0/4 = \gamma_0/2$. □

4.5.2 The Davis–Kahan/Wedin step

Theorem (Wedin $\sin\theta$, applied form). Let $M, \tilde{M} = M + E \in \mathbb{R}^{n\times d}$. Let $U, \tilde{U}$ be orthonormal bases for the leading k left singular subspaces. If $\mathrm{gap}_k(M) > 0$:

$$\sin\theta_{\max}(\mathrm{col}(U), \mathrm{col}(\tilde{U})) \leq \|E\|_F/\mathrm{gap}_k(M).$$

This is the rectangular-matrix generalization of the Davis–Kahan theorem [3] (Wedin 1972 [8]; Stewart and Sun [7]).

Application. Set $M = X_t$ and $E = X_{t+1} - X_t$. For $t \geq t_0$:

$$\sin\theta_{\max}(\mathrm{col}(X_t), \mathrm{col}(X_{t+1})) \leq \varepsilon_t/\gamma.$$

Using $\sqrt{1-x} \geq 1 - x/2$:

$$\cos\theta(\mathrm{col}(X_t), \mathrm{col}(X_{t+1})) \geq 1 - \varepsilon_t^2/(2\gamma^2) \geq 1 - C_3 \cdot e^{-2I_{t-1}}.$$

This establishes Theorem 3.1(i). □

4.6 Step 5: Dimensional reduction

Lemma 4.6 (Rank of the embedding perturbation)**.**

(a) *Each single-edge edit contributes a rank-≤-2 perturbation to* $X^e(G)$.

(b) *Therefore:* $\mathrm{rank}(X_{t+1} - X_t) \leq 2 \cdot \mathrm{d}_{\mathrm{edit}}(G_t, G_{t+1}) \leq 4C_2(K-1)e^{-I_{t-1}}$.

Proof. For e_R: $X^{e_R}(G) = A(G)M$. Adding edge (u, v) changes A by a rank-2 update, so $\Delta X = \Delta A \cdot M$ with $\mathrm{rank}(\Delta A) \leq 2$. For e_L: adding edge (u, v) perturbs the Laplacian by a rank-1 matrix $(e_u - e_v)(e_u - e_v)^T$. The bound rank $\leq 2d$ per single-edge edit holds for the d-eigenvector embedding. □

The participation ratio bound. By Lemma 4.6(b), for t sufficiently large, $\mathrm{PR}(M) \leq \mathrm{rank}(M)/n$, so:

$$\mathrm{PR}(X_{t+1} - X_t) \leq 4C_2(K-1) \cdot e^{-I_{t-1}}/n = C_4 \cdot K \cdot e^{-I_{t-1}}/n.$$

This establishes Theorem 3.1(ii). □

4.7 Proof of Corollary 3.2

Let $[V_t] \in \mathrm{Gr}(k, d)$ denote the Grassmannian point corresponding to $\mathrm{col}(X_t)$. Define $\delta_s = d_{\mathrm{Gr}}([V_s], [V_{s+1}])$. By the Wedin bound, $\delta_s \leq \sqrt{k} \cdot \varepsilon_s/\gamma \leq A \cdot e^{-I_{s-1}}$ where $A = 2\sqrt{k} \cdot C_1C_2(K-1)/\gamma$. Since I_s is non-decreasing, δ_s is non-increasing for $s \geq t_0$.

Define the tail sum $D_t = \sum_{s=t}^{T-1} \delta_s$. By the triangle inequality, $d_{\mathrm{Gr}}([V_t], [V_T]) \leq D_t$. The tail sum is strictly decreasing: $D_t - D_{t+1} = \delta_t > 0$. The cosine-to-final c_t satisfies $c_t \geq 1 - D_t^2/k$, and since D_t is strictly decreasing, the lower bound is strictly increasing for $t \geq t_0$.

Bounding discordant pairs conservatively: each of the t_0 warm-up indices can form a discordant pair with each of the $T - t_0$ post-warm-up indices, giving at most $t_0 \cdot T$ discordant pairs out of $T(T-1)/2$ total. Thus:

$$\tau_{\mathrm{to_final}} \geq 1 - 2t_0/(T-1).$$

For $T > 4t_0 + 1$, this gives $\tau_{\mathrm{to_final}} > 0.5$. □

5 Spectral Gap Obstruction (Partial Converse)

Theorem 5.1 (Spectral gap obstruction)**.** *Let (G_0, R, T) be a graph dynamical system. Denote by $\lambda_1(t) \leq \lambda_2(t) \leq \cdots \leq \lambda_n(t)$ the eigenvalues of the graph Laplacian $L(G_t)$. If there exists a subsequence $t_1 < t_2 < \cdots < t_m$ with $m > T/3$ such that $\lambda_2(t_{i+1}) < \lambda_2(t_i) - \delta$ for some $\delta > 0$, then (G_0, R, T) is not I^+ under the Laplacian embedding e_L, provided the following non-degeneracy conditions hold:*

(ND) *For each decrease step t_i, the eigenvalue $\lambda_2(t_i)$ is simple (multiplicity 1), and $\lambda_3(t_i) - \lambda_2(t_i) \geq g > 0$.*

(ND') *The eigenvalue gap at the embedding boundary satisfies $\lambda_{d+2}(t) - \lambda_{d+1}(t) \geq g' > 0$ for all t.*

Proof. The proof has three stages.

Stage 1 (Eigenvalue decrease → subspace rotation). At a decrease step t_i where $\lambda_2(t_{i+1}) < \lambda_2(t_i) - \delta$, the d-dimensional column space $V_i = \mathrm{col}(X_{t_i}^{e_L})$ is the eigenspace of the d smallest non-trivial Laplacian eigenvalues. Under (ND'), this eigenspace is separated from

its complement by gap g'. The eigenvalue λ_2 shifts by more than δ, so by the eigenspace perturbation bound [7]:

$$\sin\theta_{\max}(V_i, V_{i+1}) \geq \frac{\delta}{\text{spread}_{d+2}}$$

where $\text{spread}_{d+2} = \max_t(\lambda_{d+2}(t) - \lambda_2(t))$. Define $\eta_d = \delta/\text{spread}_{d+2} > 0$.

Stage 2 (Path length exceeds chord length). The $m > T/3$ decrease steps contribute total path length $\geq m \cdot \arcsin(\eta_d)$ on $\text{Gr}(d, n)$. The diameter of $\text{Gr}(d, n)$ is $\pi/2$. For T sufficiently large that $(T/3) \cdot \arcsin(\eta_d) > \pi$, the path must reverse direction: the cosine-to-final c_t cannot be monotonically increasing.

Stage 3 (Reversals kill $\tau_{\text{to_final}}$). Each direction reversal produces a discordant pair. The minimum number of reversals is $\geq T \arcsin(\eta_d)/(3\pi) - 1$. Once the number of discordant pairs exceeds $T/4$, the Kendall τ satisfies $\tau_{\text{to_final}} < 0.5 = \tau^*$. The reversal count exceeds $T/4$ for $T > 3\pi/(4\arcsin(\eta_d)) + 3$, a finite constant. □

Remark 5.2. Conditions (ND) and (ND') are generically satisfied. For specific rule families (e.g., tree-growing rules with high multiplicity at $\lambda = 1$), (ND) may fail at position 2 but hold at the cluster level, as in [6].

Remark 5.3 (Scope under the at-least-one rule). Theorem 5.1 shows non-I-positivity under e_L only. Under Definition 2.5's at-least-one rule, the system could still be I^+ via e_R. If both obstructions fire simultaneously, the system is I^-.

6 Computational Illustrations

6.1 Example A: Hidden Markov Model (I^+ by construction)

A 3-state HMM with known transition and emission matrices. The encoding ϕ maps the posterior to graph weights; faithfulness and smoothness hold because edge weights are linear in the posterior. Theorem 3.1 predicts $\tau_{\text{to_final}} = 1$, confirmed numerically.

6.2 Example B: Contraction mapping (convergent, non-Bayesian, I^-)

A graph on $n = 20$ nodes. Rewrite rule: at each step, remove the edge with maximum betweenness centrality and add an edge between the two nodes closest in graph distance. This contracts the graph toward a path or tree. Under the canonical specification, Example B classifies as I^- with score 2/4. The per-seed τ values are 0.50, 0.44, 0.00, and 0.50. The contraction mapping reaches its fixed point within 3–5 steps.

Diagnostic finding. Grassmannian trajectory analysis reveals a discriminating structural feature: contraction trajectories have high *straightness* (ratio of direct Grassmannian distance to total path length $\approx$ 0.5–0.8), while genuinely I^+ rules show low straightness ($\approx$ 0.34–0.37). This motivates a Grassmannian straightness gate for future predicate revisions (Section 7).

6.3 Example C: Cellular automaton (I^-)

A 1D cellular automaton (Rule 110) lifted to a graph. Complex, Turing-complete dynamics. The spectral gap obstruction (Theorem 5.1) fires: the spectral gap oscillates irregularly.

6.4 Example D: Random rewiring (null model, I^-)

Start from $G(20, 0.3)$. At each step, perform one degree-preserving random edge swap. No progressive alignment ($\tau_{\text{to_final}} \approx 0$). This is the null model against which τ^* is calibrated.

6.5 Summary

Example	Bayesian?	Convergent?	I-class	Gap obstruction?
A (HMM)	Yes	Yes	I^+	No (gap increases)
B (Contraction)	No	Yes	I^-	Possibly not
C (CA Rule 110)	No	No	I^-	Yes
D (Random rewiring)	No	No	I^-	Yes

Table 1: Summary of the four discrimination examples.

Example B is the critical test of predicate specificity. The I-predicate correctly classifies it as I^- under the canonical specification, though the margin is thin and motivates the straightness gate discussed in Section 7.

7 Discussion

7.1 Architecture independence

The forward theorem holds for any system satisfying Definition 2.6 regardless of graph topology or update mechanism. This is genuine architecture independence: the same geometric signatures emerge whether the "hardware" is a transformer, a Mamba block, or an HMM encoded as a graph. However, we do not claim the converse—that any I^+ system is necessarily Bayesian.

7.2 The I-predicate as a standalone contribution

The I-predicate provides a formal, measurable test for a specific class of dynamical behavior; built-in robustness via independent embeddings with an at-least-one rule and anti-convergence guard; and a continuous relaxation (the I-score) suitable for temporal profiling. The forward theorem validates the predicate in one direction; the discrimination examples probe its specificity.

7.3 Limitations

Finite Θ. Extension to continuous Θ requires Bernstein–von Mises type arguments.

The encoding ϕ. The smoothness condition on ϕ is the least natural part of Definition 2.6. Graph encodings of belief states may be discontinuous when the topology changes discretely.

Laplacian sign ambiguity. The eigenvector embedding e_L has a sign/basis ambiguity for graphs with repeated Laplacian eigenvalues. The at-least-one rule is specifically designed for this; e_R carries the classification alone in such cases.

7.4 The Grassmannian straightness signal

Example B revealed that the I-predicate's discrimination between Bayesian convergence and generic contraction is thinner than desired. The straightness of the Grassmannian trajectory, defined as

$$S = d_{\mathrm{Gr}}([V_0], [V_T]) \Big/ \sum_{s=0}^{T-1} d_{\mathrm{Gr}}([V_s], [V_{s+1}]),$$

discriminates: contraction mappings follow near-geodesic paths ($S \approx 0.5$–0.8) while genuinely inference-like systems follow winding paths ($S \approx 0.34$–0.37). Formalizing a straightness gate is future work.

7.5 Open problems

1. **Tight characterization of I-positivity.** What is the exact class of systems that satisfy the I-predicate?
2. **Continuous Θ.** Extend Theorem 3.1 to continuous parameter spaces.
3. **Rate optimality.** Is the information-gain bound in Theorem 3.1(i) tight?
4. **Computational complexity.** Given a graph rewrite rule R, how hard is it to determine whether (G_0, R, T) admits a Bayesian certificate?
5. **Dimensional reduction criterion.** Can strict monotonicity be relaxed to a Kendall τ formulation?
6. **Straightness bound.** Prove that Bayesian systems produce Grassmannian trajectories with $S \leq 1 - f(K, C_1, C_2)$.

A Verification of E1–E3 for Standard Embeddings

A.1 Laplacian eigenvectors

E1. Adding or removing a single edge (u, v) perturbs the Laplacian by $\Delta L = \pm(e_u - e_v)(e_u - e_v)^T$, a rank-1 update with $\|\Delta L\|_2 \leq 2$. By Weyl's inequality [9], each eigenvalue shifts by at most 2. By the Davis–Kahan theorem [3], each eigenvector shifts by at most $2/\min_j |\lambda_j - \lambda_{j'}|$. Summing over d eigenvectors gives $\|\Delta X^{e_L}\|_F \leq C_1$.

E2. Co-spectral graphs with identical eigenvector structure are rare and resolved by combining e_L with e_R.

E3. The singular values of $X^{e_L}(G)$ are all 1 (orthonormal eigenvectors). Spectral information is carried by which eigenvalues are selected.

A.2 Random projection

E1. $X^{e_R}(G) = A(G)M$. A single-edge change gives $\|\Delta X^{e_R}\|_F \leq \sqrt{2} \cdot \|M\|_2 = O(\sqrt{n})$.

E2. $A \neq A'$ implies $AM \neq A'M$ with probability 1 over M.

E3. Singular values of AM concentrate around those of A by the Johnson–Lindenstrauss guarantee [5].

B Null Model Specification

Null model N1 (Degree-preserving random dynamics). Given G_0 and T, generate a trajectory by performing one degree-preserving random edge swap at each step. The threshold $\tau^* = 0.5$ is set above the observed maximum $\tau_{\text{to_final}}$ of 0.37 across 20 independent ER trajectories.

C Computational Details

All examples: $T = 30$, embedding dimension $d = 5$, compression via zlib level 9 on JSON-serialized edge lists. Initial graphs: K_1, K_2, K_3, P_3, each bootstrapped to $n = 20$ for Example B. Reproducible with numpy seed 42.

D Summary of Constants

Symbol	Definition	Depends on
$K = \lvert\Theta\rvert$	Parameter space size	Model
C_1	Embedding continuity (E1)	max degree, e
C_2	Encoding smoothness (Definition 2.6)	ϕ
$\gamma_0 = \text{gap}_k(X^e(G_\infty))$	Limit singular value gap	G_∞, e, k
$\gamma = \gamma_0/2$	Working gap constant	γ_0
$C_3 = 2C_1^2C_2^2(K-1)^2/\gamma^2$	Progressive alignment bound	C_1, C_2, K, γ
$C_4 = 4C_2$	Dimensional reduction bound	C_2
D_0	Minimum discrimination rate	Identifiability
$t_0 \leq (4C_1C_2(K-1)/\gamma_0)^{1/D_0}$	Warm-up time	$C_1, C_2, K, \gamma_0, D_0$
$T > 4t_0 + 1$	I-positivity threshold (Corollary 3.4)	t_0

Table 2: Summary of constants used in the forward theorem.

The minimum discrimination rate D_0 is the standard Bayesian identifiability parameter. It is positive iff the model is identifiable.

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